Program: B. Tech. (CSE)	Course Title: Artificial Intelligence			Course Code: CS312
Type of Course: Program Core	**Prerequisites:** Data Structures and Algorithms (CS121), Discrete Mathematics (CS124) , Probability and Statistics (MT211)			**Total Contact Hours:** 40L + 20P = 60
Year/Semester: 3/Odd	**Lecture Hrs/Week:** 3	**Tutorial Hrs/Week:** 0	**Practical Hrs/Week:** 2	**Credits:** 4

Learning Objective:

The course aims to provide a broad introduction to the field of artificial intelligence, its philosophical foundations, success and failures. Intelligent entities and their importance are explored for different domains in AI. The topics include problem solving in AI, knowledge representation, reasoning, learning, and planning to understand the ways to make computers intelligent in problem solving. The course will help the students in formulating the solutions to different categories of problems in artificial intelligence through case studies in different fields. Furthermore, the students will be able to approach to solve the different Engineering problems by implementing the concepts learned in the course.

Course outcomes (COs):

On completion of this course, the students will be able to:		Bloom's Level
CO-1	**Relate** the role of agents in identifying the problems that are amenable to solutions by AI methods.	2
CO-2	**Apply** the various knowledge representation techniques to represent contextual and semantic-oriented information efficiently.	3
CO-3	**Analyze** probabilistic solutions to implement decision-making in different fields of AI.	4
CO-4	**Evaluate** the significance of planning and learning in the development of intelligent systems.	5
CO-5	**Construct** the solution methodologies for the real-world problems by identifying the applications and challenges in AI.	3

Course Topics	Lecture Hours	Associated CO
UNIT – I (Introduction)	**10**	**CO1**
1.1 Introduction to AI, history and philosophical foundations, Ethics in AI, Agents and Environment: rationality, agent and environment types and their significance in AI.	2	
1.2 Problem solving in AI, well-defined problem, Examples of problems, Uninformed Search: breadth first search, depth first search, depth limited	4	

search, iterative deepening search, uniform cost search; Informed Search: heuristic function, hill climbing, best first search, greedy search, A* and AO* search.		
1.3 Constraint satisfaction problem (CSP) and constraint optimization problem (COP), evaluation function; Game playing: game tree, 2-player zero-sum game, Mini-Max search, alpha-beta pruning, Expectimax.	**4**	
UNIT – II (Knowledge Representation)	**10**	**CO2**
2.1 Knowledge Representation: mapping between facts and representation, approaches and techniques of knowledge representation, Knowledge based agents, Predicate logic, unification, and resolution.	**3**	
2.2 Rule based system, forward reasoning and conflict resolution, backward reasoning, and its use; Introduction to Fuzzy logic and Fuzzy system: membership function, dilation, concentration, fuzzification, de-fuzzification.	**3**	
2.3 Structured knowledge representation: semantic networks – slot and inheritance, frames, scripts, conceptual dependency; introduction to ontology in AI, description logic.	**4**	
UNIT – III (Probabilistic Reasoning)	**10**	**CO3**
3.1 Handling uncertainty: sources of uncertainty, belief state, rational decision, utility theory, decision theory; Probabilistic inference: conditional, marginal, joint distribution; Bayes' rule, Naïve Bayes and its limitations, Bayesian Belief Network (BBN), Inference with BBN, factor graphs.	**5**	
3.2 Markov Model: Markov Chain, Markov Decision Process (MDP): value iteration, policy iteration, Partially Observable MDP: model-based and model-free approaches, Hidden Markov Model: introductory concepts, filtering, smoothing.	**5**	
UNIT-IV (Learning and Planning)	**7**	**CO4**
4.1 Learning: introductory concepts, learning model, supervised and unsupervised learning; Artificial Neural Network: characteristics, types of ANN architectures, activation function; introduction to reinforcement learning: Monte Carlo methods, Q-learning, SARSA.	**4**	
4.2 Planning: the planning problem, planning with state space search, planning graph, planning example – total order planning, partial-order planning, STRIPS, PDDL.	**3**	
UNIT-V (Applications)	**3**	**CO5**
5.1 Introduction to AI languages; Case Studies of AI applications to real-world problems.	**3**	

Laboratory	Lab Hours	Associated CO
List of Experiments	**20**	
1. Visualize the agent-based solution for different use cases.	**2**	**CO1, CO5**
2. Implement the solution to the problems using uninformed search techniques.	**2**	**CO1**
3. Implement the solution to the problems using informed search techniques.	**2**	**CO1**
4. Use the constraint satisfaction technique to provide solutions in the domain of game playing.	**2**	**CO1**
5. Implementation of different knowledge representation schemes for various use cases.	**2**	**CO2**
6. Apply the concepts of unification and resolution for real-world problems.	**2**	**CO2**
7. Use Naïve Bayes classifier to solve the problems of Bayesian Belief Network.	**2**	**CO3**
8. Illustrate the importance of Markov models through implementation for different case studies.	**2**	**CO3**
9. Implementation of planning solutions to the problems using different techniques in planning.	**2**	**CO4**
10. Implement the different learning algorithms for the different sets of use cases.	**2**	**CO4**

Tools and Platforms
1. **Traditional Platforms:** LISP, OPS5, Prolog
2. **Additional Platforms:** As per the current practices, the AI lab can be conducted using tools and platforms such as Python, MATLAB, Clips, NetLogo, Keras, Jade, Spade, Tensorflow, Caffe, etc.

Textbook References:
Textbook:
1. Stuart Russell and Peter Norvig, *Artificial Intelligence – A Modern Approach*, 4th Edition, Pearson Education, 2021.
2. Elaine Rich and Kevin Knight, *Artificial Intelligence*, 3rd Edition, Tata McGraw Hill, 2017.

Reference books:
1. Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd Edition, MIT Press, 2018.

2. Dan W. Patterson, *Introduction to Artificial Intelligence and Expert Systems*, Prentice Hall of India.
3. Nils J. Nilsson, *The Quest for Artificial Intelligence*, Cambridge University Press. 2009.
4. P.H. Winston, *Artificial Intelligence*, 3rd Edition, Addison-Wesley Publishing Company, 1993.
5. William F. Clocksin and Christopher S. Mellish, *Programming in PROLOG*, 5th Edition, Springer.
6. Deepak Khemani, *A First Course in Artificial Intelligence*, 1st Edition, McGraw-Hill Education, 2017.
7. George F. Lugar, *Artificial Intelligence – Structures and Strategies for Complex Problem Solving*, 6th Edition, Pearson Education Limited, 2008.
8. Ben Coppin, *Artificial Intelligence Illuminated*, 1st Edition, Jones and Bartlett Publishers, 2004.
9. Nils J. Nilsson, *Principles of Artificial Intelligence*, 1st Edition, Springer, 2014.
10. Uri Wilensky and William Rand, *An Introduction to Agent-based Modeling*, MIT Press, 2015.

Evaluation Method		
Item	**Weightage (%)**	**Associated CO**
Quiz	15	CO1, CO3
Mid Term	20	CO1, CO2
End term	40	CO1, CO2, CO3, CO4, CO5
Lab	25	CO1, CO2, CO3, CO4, CO5

*Please note, as per the existing institute's attendance policy the student should have a minimum of 75% attendance. Students who fail to attend a minimum of 75% lectures will be debarred from the End Term/Final/Comprehensive examination.

CO and PO Correlation Matrix (CSE)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	2	1	2	1							1	3	1	1
CO 2	3	2	2	1	2								3	2	2
CO 3	3	3	3	3	1								3	2	2
CO 4	3	2	3	2	2								3	3	3
CO 5	3	3	3	3	3	1	1	1	1	1		2	2	3	3

CO and PO Correlation Matrix (B.Tech. – M.Tech. Integrated Dual Degree (CSE))

CO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO1	PO1	PO1	PSO	PSO	PSO

	1	2	3	4	5	6	7	8	9	0	1	2	1	2	3
CO 1	3	2	1	2	1							1	3	1	1
CO 2	3	2	2	1	2								3	2	2
CO 3	3	3	3	3	1								3	2	2
CO 4	3	2	3	2	2								3	3	3
CO 5	3	3	3	3	3	1	1	1	1	1		2	2	3	3

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Updated By: Saurabh Kumar, Poulami Dalapati
Approved By: